Volumes Of Revolution

Chapter Overview

Choose the axis

About the x-axis: integrate πy^2 with respect to x.

About the y-axis: integrate πx^2 with respect to y.

Choose the method

Convert limits when using a parameter.

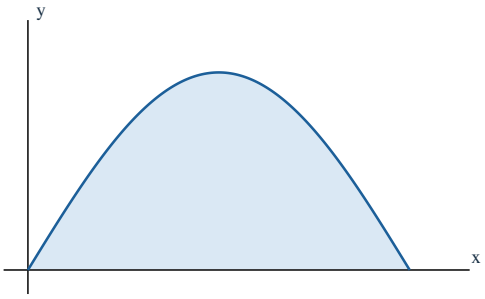
Subtract inner volumes for hollow regions.

Apply cubic scale factors to volumes in models.

Volumes of Revolution with harder integration

Solution 3

The region under $y = \sin 2x$, from $x = 0$ to $x = \pi/2$, is rotated about the x-axis. Find its volume.



Disc method

A thin disc has radius y and thickness dx.

$$dV = \pi y^2 dx, \qquad V = \pi \int_a^b y^2 dx$$

Evaluate

$$V = \pi \int_0^{\pi/2} \sin^2 2x \, dx$$

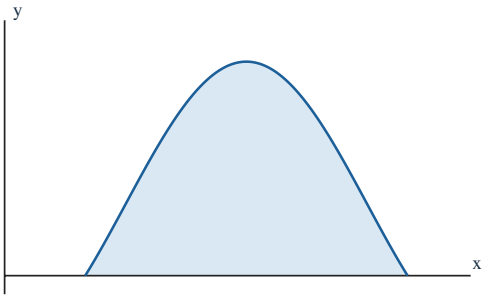
$$= \pi [x/2 - \sin 4x/8]_0^{\pi/2} = \frac{\pi^2}{4}$$

Test Your Understanding

Solution 4

Edexcel C4 January 2013, Q6

The region between $y = 1 - 2\cos x$ and the x-axis, from its first to second positive roots, is rotated about the x-axis. Find the exact volume.



Limits

$$A = \pi/3, \quad B = 5\pi/3$$

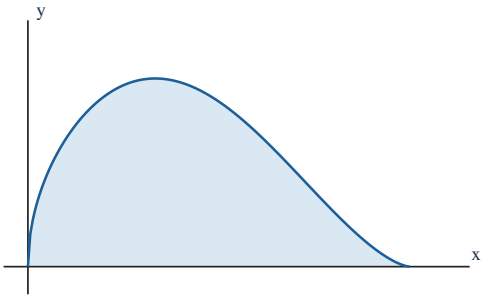
Volume

$$V = \pi \int_{\pi/3}^{5\pi/3} (1 - 2\cos x)^2 dx$$

$$= \pi [3x - 4\sin x + \sin 2x]_{\pi/3}^{5\pi/3}$$

$$= 4\pi^2 + 3\pi\sqrt{3}$$

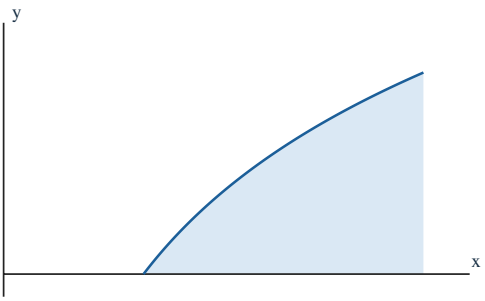
2. Rotate the region under $y = \cos x\sqrt{\sin 2x}$, $0 \le x \le \pi/2$, about the x-axis. Find its volume.



2

$$V = \pi \int_0^{\pi/2} \cos^2 x \sin 2x \, dx$$
$$= 2\pi \int_0^{\pi/2} \sin x \cos^3 x \, dx$$
$$= \pi\left[-\tfrac{1}{2}\cos^4 x\right]_0^{\pi/2} = \frac{\pi}{2}$$

3. The region bounded by $y = \ln x$, the x-axis and $x = 3$ is rotated about the x-axis. Find its exact volume.



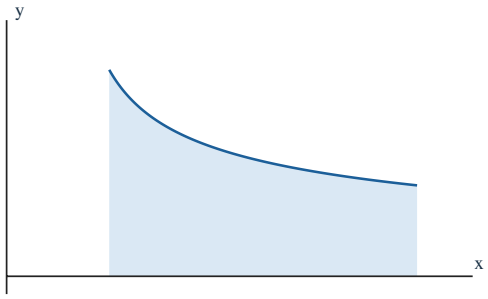
3

The curve meets the axis at $x = 1$.

Integrate by parts twice:

$$V = \pi \int_1^3 (\ln x)^2 dx$$
$$= \pi\left[x(\ln x)^2 - 2x\ln x + 2x\right]_1^3$$
$$= \pi(3(\ln 3)^2 - 6\ln 3 + 4)$$

5. A curve satisfies $y^2 = (4x+3)/((x+2)(2x-1))$. Find the volume generated by rotating the region under its positive branch, $1 \leq x \leq 4$, about the x-axis.



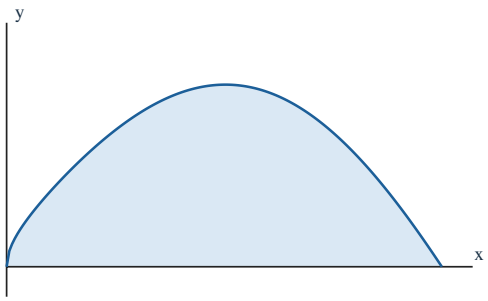
Split

$$\frac{4x+3}{(x+2)(2x-1)} = \frac{1}{x+2} + \frac{2}{2x-1}$$

Volume

$$V = \pi[\ln(x+2) + \ln(2x-1)]_1^4 = \pi \ln 14$$

6. The curve $2y^2 = x\sin x + x$ forms a loop from the origin to its first positive x-axis intersection. Find the volume formed by rotating the loop about the x-axis.



Intersection

$$\sin x = -1 \Rightarrow A = (3\pi/2, 0)$$

Volume

$$V = \frac{\pi}{2}\int_0^{3\pi/2} (x\sin x + x)dx$$

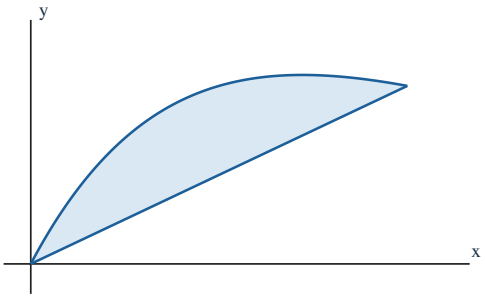
$$= \frac{\pi}{2}[-x\cos x + \sin x + x^2/2]_0^{3\pi/2}$$

$$= \frac{\pi}{2}\left(\frac{9\pi^2}{8} - 1\right)$$

Exercise 4A - Core Pure Year 2 Pages 78-80

Solution 9

8. The positive region between $y = xe^{-x}$ and $y = x/4$ is rotated about the x-axis. Find its volume to 3 significant figures.



Intersections

$xe^{-x} = x/4 \Rightarrow x = 0,\ \ln 4$

Washer method

$$V = \pi \int_0^{\ln 4} (x^2 e^{-2x} - x^2/16)dx$$

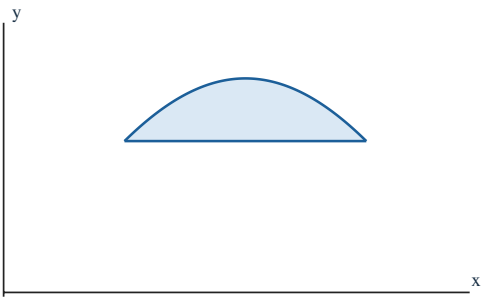
$$= \pi[-e^{-2x}(x^2/2 + x/2 + 1/4) - x^3/48]_0^{\ln 4}$$

$$\approx 0.237$$

Challenge 4A - Core Pure Year 2 Pages 78-80

Solution 10

Challenge. Rotate the region between $y = \sin x$ and $y = 1/\sqrt{2}$, for $\pi/4 \leq x \leq 3\pi/4$, about the line $y = 1/\sqrt{2}$. Find the exact volume.



Shift the axis

The radius is the distance above the rotation axis:

$$V = \pi \int_{\pi/4}^{3\pi/4} (\sin x - 1/\sqrt{2})^2 dx$$

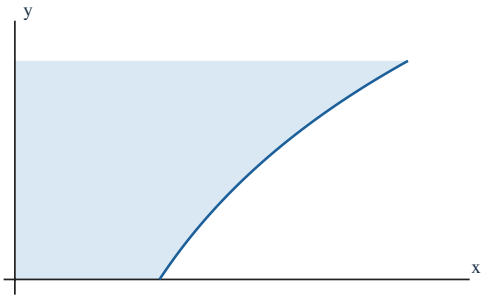
$$= \pi[x - \sin 2x/4 + \sqrt{2}\cos x]_{\pi/4}^{3\pi/4}$$

$$= \frac{\pi}{2}(\pi - 3)$$

Revolving around the y-axis

Solution 11

The region between the y-axis and $y = 4\ln x - 1$, for $0 \le y \le 4$, rotates about the y-axis. Find its volume.



Formula

$$V = \pi \int_c^d x^2 dy, \qquad x = e^{(y+1)/4}$$

Evaluate

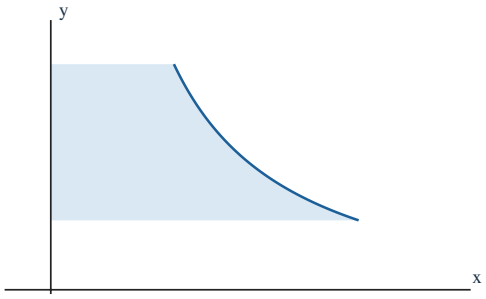
$$V = \pi \int_0^4 e^{(y+1)/2} dy$$

$$= 2\pi \left[e^{(y+1)/2}\right]_0^4 = 2\pi\sqrt{e}(e^2 - 1)$$

Exercise 4B - Core Pure Year 2 Pages 81-83

Solution 12

3. The region between $x = 1/(2y+1)$, the y-axis and $y = 1, b$, where $b > 1$, rotates about the y-axis. Its volume is $\pi/10$. Find b.



Integral

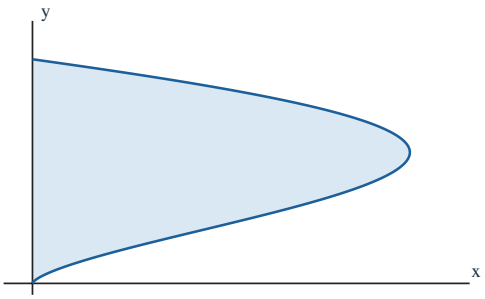
$$V = \pi \int_1^b (2y+1)^{-2} dy = \frac{\pi}{2}\left(\frac{1}{3} - \frac{1}{2b+1}\right)$$

Solve

$$\frac{1}{3} - \frac{1}{2b+1} = \frac{1}{5} \Rightarrow 2b + 1 = \frac{15}{2}$$

$$b = \frac{13}{4}$$

4. The first lobe of $x = \sqrt{y}\sin y$, $y \geq 0$, is bounded by the curve and the y-axis. Find its upper endpoint and the volume generated about the y-axis.



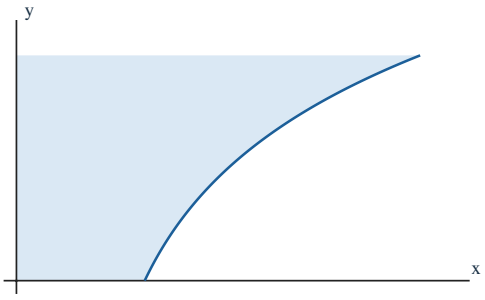
Endpoint

$\sin y = 0 \Rightarrow b = \pi$

Volume

$$V = \pi \int_0^{\pi} y \sin^2 y \, dy$$
$$= \frac{\pi}{2}[y^2/2 - y\sin 2y/2 - \cos 2y/4]_0^{\pi}$$
$$= \frac{\pi^3}{4}$$

5. The region between $y = 3\ln(x-1)$, the y-axis, $y = 0$ and $y = 5$ is rotated about the y-axis. Find its exact volume.



Rearrange

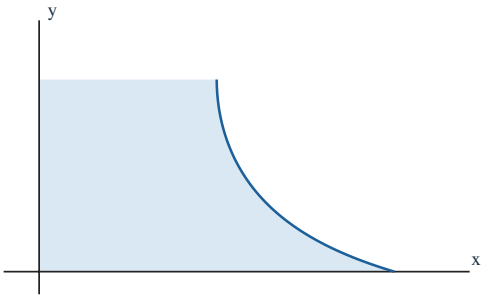
$x = 1 + e^{y/3}$

Volume

$$V = \pi \int_0^5 (1 + e^{y/3})^2 dy$$
$$= \pi[y + 6e^{y/3} + \tfrac{3}{2}e^{2y/3}]_0^5$$
$$= \pi(\tfrac{3}{2}e^{10/3} + 6e^{5/3} - \tfrac{5}{2})$$

6a. Write $\cos y + \sqrt{3}\sin y$ as $R\cos(y-\alpha)$.

b. Find the volume about the y-axis of the region between that axis and $x = 1/(\cos y + \sqrt{3}\sin y)$, for $0 \le y \le \pi/3$.



a

$$R = 2, \quad \alpha = \pi/3$$

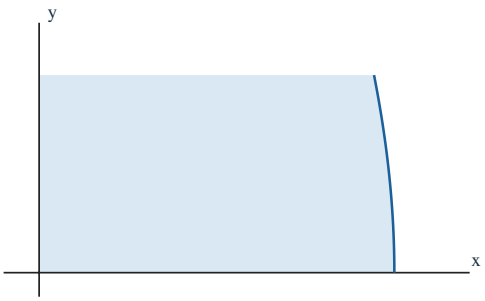
b

$$V = \frac{\pi}{4}\int_0^{\pi/3} \sec^2(y-\pi/3)dy$$

$$= \frac{\pi}{4}[\tan(y-\pi/3)]_0^{\pi/3} = \frac{\pi\sqrt{3}}{4}$$

7a. Use $u = 2^y$ to evaluate $\int_0^1 2^y/(2^y+1)^2 dy$.

b. Hence find the volume generated about the y-axis by $x = \sqrt{2^y}/(2^y+1)$, between $y = 0$ and 1.



a

$$dy = \frac{du}{u\ln 2}, \quad I = \frac{1}{\ln 2}\int_1^2 \frac{du}{(u+1)^2}$$

$$= \frac{1}{\ln 2}\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{1}{6\ln 2}$$

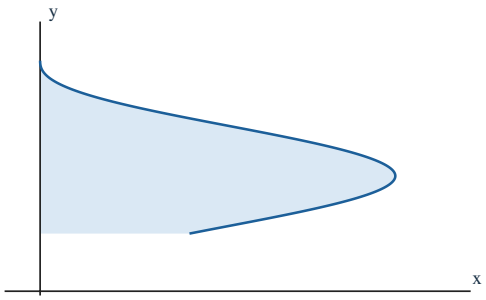
b

$$V = \pi I = \frac{\pi}{6\ln 2}$$

Exercise 4B - Core Pure Year 2 Pages 81-83 — Solution 17

8a. Prove $\sin^5\theta = (10\sin\theta - 5\sin 3\theta + \sin 5\theta)/16$.

b. Rotate the region between $x = \sin^2 y\sqrt{\sin y}$ and the y-axis, $\pi/4 \le y \le \pi$. Find the exact volume.



a

Writing $s = \sin\theta$, use $\sin 3\theta = 3s - 4s^3$ and $\sin 5\theta = 5s - 20s^3 + 16s^5$; eliminate s^3 and rearrange.

b

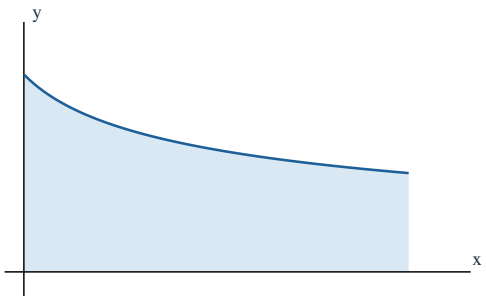
$$V = \frac{\pi}{16}[-10\cos y + \tfrac{5}{3}\cos 3y - \tfrac{1}{5}\cos 5y]_{\pi/4}^{\pi}$$

$$= \frac{\pi}{15}\left(8 + \frac{43\sqrt{2}}{8}\right)$$

Volumes of revolution for parametric curves — Solution 18

Parametric curves

$x = t(1+t)$, $y = 1/(1+t)$, $t \ge 0$. Rotate the region under the curve from $x = 0$ to 2 about the x-axis.



Change variable

$$V = \pi\int y(t)^2\frac{dx}{dt}dt$$

The parameter limits are 0 and 1; $dx/dt = 1 + 2t$.

Evaluate

$$V = \pi\int_0^1 \frac{1+2t}{(1+t)^2}dt$$

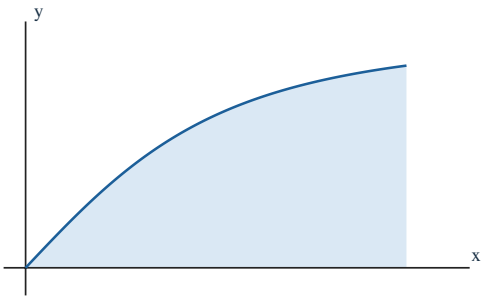
$$= \pi[2\ln(1+t) + (1+t)^{-1}]_0^1 = \pi(2\ln 2 - \tfrac{1}{2})$$

Test Your Understanding

Solution 19

Edexcel C4 June 2011, Q7

A curve has $x = \tan\theta,\ y = \sin\theta$. Rotate the region under it, $0 \le x \le \sqrt{3}$, about the x-axis. Find the exact volume.



Limits

$0 \le \theta \le \pi/3, \quad dx/d\theta = \sec^2\theta$

Volume

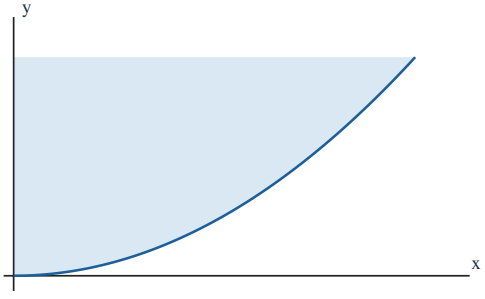
$$V = \pi \int_0^{\pi/3} \sin^2\theta \sec^2\theta \, d\theta$$

$$= \pi[\tan\theta - \theta]_0^{\pi/3} = \pi\sqrt{3} - \frac{\pi^2}{3}$$

Exercise 4C - Core Pure Year 2 Pages 84-87

Solution 20

6. The curve $x = 2t,\ y = t^2$, $-2 \le t \le 2$, meets the horizontal line $y = a$ at its endpoints. Find a and the volume of the bounded region rotated about the y-axis.



Set up

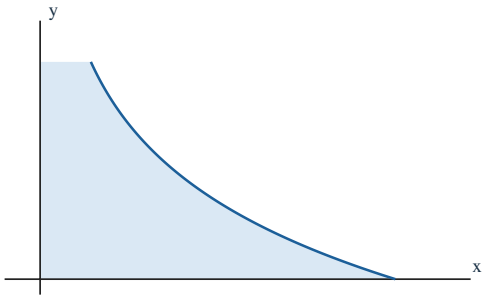
$a = 4, \quad V = \pi \int x^2 dy$

Use only the right-hand half: rotating it already generates the whole solid.

Volume

$$V = \pi \int_0^2 (2t)^2(2t)dt = 8\pi[t^4/4]_0^2 = 32\pi$$

8. A curve has $x = 1/(2t),\ y = \ln(2t),\ t \geq 1/2$. The region between it, the coordinate axes and $y = a$ rotates about the y-axis. Its volume is $24\pi/49$. Find a.



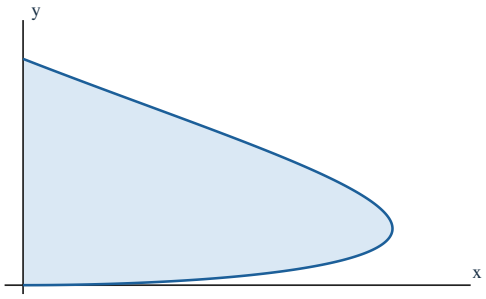
Integral

$$x = e^{-y}, \quad V = \pi \int_0^a e^{-2y} dy = \frac{\pi}{2}(1 - e^{-2a})$$

Solve

$$e^{-2a} = \frac{1}{49} \Rightarrow a = \ln 7$$

9. The curve $x = 2\sin t,\ y = t^2,\ 0 \leq t \leq \pi$, encloses a region with the y-axis. Find the volume when it rotates about that axis.



Set up

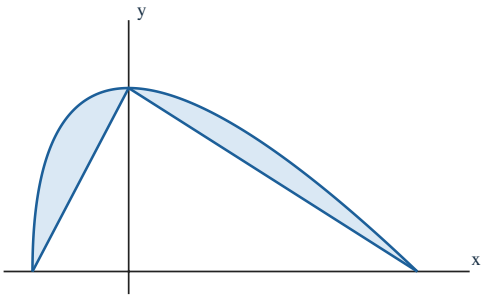
$$V = \pi \int_0^\pi (2\sin t)^2 (2t) dt = 4\pi \int_0^\pi t(1 - \cos 2t) dt$$

Integrate

$$\int t\cos 2t\, dt = \tfrac{t}{2}\sin 2t + \tfrac{1}{4}\cos 2t$$

$$V = 4\pi[t^2/2 - t\sin 2t/2 - \cos 2t/4]_0^\pi = 2\pi^3$$

10. The curve $x = t^2 - 2t,\ y = 1 - t^2,$ $-1 \le t \le 1$, intersects the axes at P, Q, S. Rotate the region between the curve and chords PQ, QS about the x-axis. Find its volume.



Intercepts

$P = (-1, 0),\ Q = (0, 1),\ S = (3, 0)$

Subtract cones

$$V_{\text{curve}} = \pi \int_1^{-1} (1 - t^2)^2 (2t - 2) dt = \frac{32\pi}{15}$$

$$V_{\text{cones}} = \frac{\pi}{3} + \pi = \frac{4\pi}{3}$$

$$V = \frac{32\pi}{15} - \frac{4\pi}{3} = \frac{4\pi}{5}$$

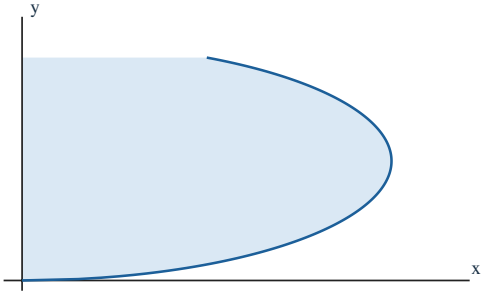
Modelling with Volumes of Revolution

Solution 24

A goldfish bowl is modelled by rotating $x = 2\sin t,\ y = 2\cos t + 2,$ $\pi/6 \le t \le 11\pi/6$, about the y-axis.

a. Find the volume up to height 3 cm.

b. A similar real bowl has maximum diameter 48 cm. Find the corresponding water volume.



a

$$x^2 = 4 - (y - 2)^2 = 4y - y^2$$

$$V = \pi \int_0^3 (4y - y^2) dy = 9\pi \text{ cm}^3$$

b

The model maximum diameter is 4 cm; linear scale factor is 12.

$$V_{\text{real}} = 12^3 (9\pi) = 15552\pi \approx 48900 \text{ cm}^3$$

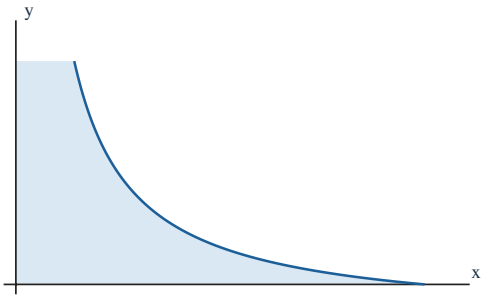
Exercise 4D - Core Pure Year 2 Pages 88-89

Solution 25

1. A volcano is modelled by rotating the region $0 \le x \le 2000/(20+y)$, $0 \le y \le 120$, about the y-axis. Units are metres.

a. Find the base diameter.

b. Estimate its volume.



a

$x(0) = 100 \Rightarrow d = 200 \text{ m}$

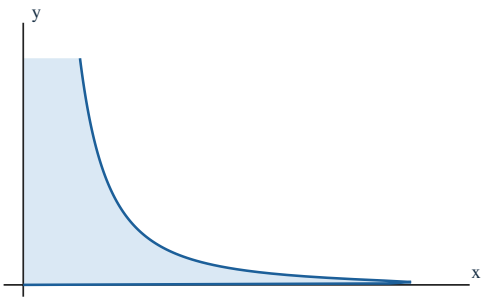
b

$$V = 4000000\pi \int_0^{120} (20+y)^{-2} dy$$
$$= 4000000\pi [-(20+y)^{-1}]_0^{120}$$
$$= \frac{1200000\pi}{7} \text{ m}^3$$

Exercise 4D - Core Pure Year 2 Pages 88-89

Solution 26

2. A vase of height 30 cm is formed by rotating $x^2 = 100y/(10y^2+1)$, $0 \le y \le 30$, about the y-axis. It is filled to height 20 cm. Find the exact water volume as $p\pi \ln q \text{ cm}^3$, for integers p, q.



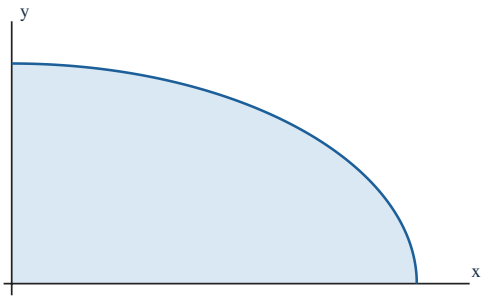
2

$$V = \pi \int_0^{20} \frac{100y}{10y^2+1} dy$$
$$= 5\pi [\ln(10y^2+1)]_0^{20} = 5\pi \ln 4001 \text{ cm}^3$$
$$p = 5, \quad q = 4001$$

Exercise 4D - Core Pure Year 2 Pages 88-89 — Solution 27

3a. Prove $\cos^3\theta = \frac{3}{4}\cos\theta + \frac{1}{4}\cos 3\theta$.

b. A domed tent is formed by rotating $x = 50\cos\theta,\ y = 30\sin\theta$, $0 \le \theta \le \pi/2$, about the y-axis. Find its volume in cubic metres.



a

$\cos 3\theta = \cos(2\theta + \theta) = 4\cos^3\theta - 3\cos\theta$

Rearrange to obtain the identity.

b

$$V = 75000\pi \int_0^{\pi/2} \cos^3\theta\, d\theta$$

$$= \frac{75000\pi}{4}\left[3\sin\theta + \tfrac{1}{3}\sin 3\theta\right]_0^{\pi/2}$$

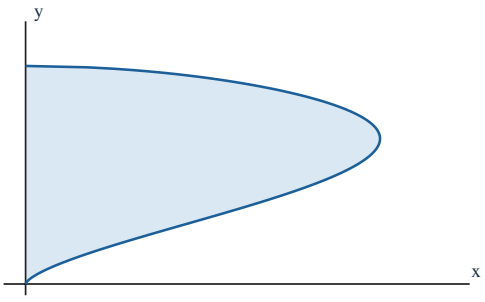
$$= 50000\pi \text{ m}^3$$

Exercise 4D - Core Pure Year 2 Pages 88-89 — Solution 28

4. A balloon model is formed by rotating $x = \sin y\sqrt{\sin 2y}$, $0 \le y \le \pi/2$, about the y-axis. Units are metres.

a. Find its volume.

b. The real balloon has height 6π metres. Find its volume.



a

$$V = 2\pi \int_0^{\pi/2} \sin^3 y\cos y\, dy$$

$$= \frac{\pi}{2}\left[\sin^4 y\right]_0^{\pi/2} = \frac{\pi}{2} \text{ m}^3$$

b

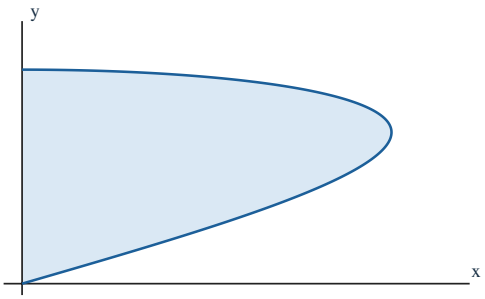
$$k = \frac{6\pi}{\pi/2} = 12, \quad V_{\text{real}} = 12^3\frac{\pi}{2} = 864\pi \text{ m}^3$$

5. A silver earring is formed by rotating $x = 2\sin 2\theta,\ y = 3\sin\theta$, $0 \le \theta \le \pi$, about the y-axis. Units are millimetres.

a. Show $x^2 = \frac{16}{81}y^2(9 - y^2)$.

b. Find the maximum number made from 300 mm^3 of silver.

c. Discuss modelling limitations.



a

$$x^2 = 16\sin^2\theta(1 - \sin^2\theta) = \frac{16}{81}y^2(9 - y^2)$$

b

$$V = \frac{16\pi}{81}\int_0^3 (9y^2 - y^4)dy = \frac{32\pi}{5}$$

$$\left\lfloor \frac{300}{32\pi/5} \right\rfloor = 14 \text{ earrings}$$

c

A patterned or hollow earring may use less silver; loss during casting may reduce the number made.